Andrew Li

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Education

Georgia Institute of Technology — Atlanta, GA
M.S. in Computer Science — *Expected May 2027*
B.S. in Computer Science, GPA: 3.74 — *Aug. 2023 – May 2026*

- **Coursework:** Computer Vision, Perception and Robotics, Human-Robot Interaction, Machine Learning, DSA
- **Activities:** Humanoid Robotics (Software Lead); RoboWrestling (Software Mentor); Flowers Invention Studio (PI)

Experience

Georgia Tech Lunar Lab — Atlanta, GA
Research Assistant — *Apr. 2026 – Present*

- Building a real-to-sim pipeline that turns a single handheld LiDAR and tri-camera capture into a photorealistic, collision-ready digital twin for robot training in large-scale indoor environments
- Adapted FAST-LIVO2 LiDAR-inertial-visual odometry (LiDAR-IMU-camera fusion) for camera pose estimation
- Implementing 3D reconstruction methods including Gaussian Splatting, NeRF, and SDF

Sandia National Laboratories — Albuquerque, NM (On-site, then Remote)
Year-Round AI for Autonomy Intern — *May 2025 – Present*

- Developed a custom C++ ROS 2 interface for Kromek GR1+ spectrometers and SparkFun GPS, then led Jackal unmanned ground vehicle (UGV) field trials for real-time radiation mapping, detailed in report SAND2025-12871R
- Developing embedded C++ firmware for a microcontroller-based drone (UAV) flight logger that captures IMU, GPS, and barometric altimeter data
- Engineered a semantic navigation stack in Isaac Sim for a Spot quadruped, pairing a local Gemma 3 vision language model (VLM) via Ollama for contextual reasoning with behavior trees for high-level planning

Projects

AGATHA: Automated Self-Driving Hazard Detection and Scenario Mining — *Oct. 2026 – Present*

- Developing a Python pipeline to mine the Waymo Open Dataset for driving hazards, including vehicle cut-ins, unexpected pedestrian behavior, and road construction
- Integrated Gemma VLMs via vLLM for local GPU inference, analyzing video and 3D object trajectories to classify incidents across 14 hazard types
- Developing a PostgreSQL database to organize incidents by hazard type and support search and video review through a Streamlit web application

HARPOR: Video to Physics-Ready Simulation Assets — *Aug. 2026 – Present*

- Building a video-to-3D pipeline that turns object videos into simulation assets with estimated physical properties
- Integrating DA3 depth estimation and MV-SAM3D reconstruction with agents that refine geometry and appearance
- Designed a VLM agent to add movable joints in Blender and repair articulation using simulation feedback in MuJoCo
- Built a FastAPI web UI for video uploads, interactive 3D previews, and live reconstruction progress tracking

Basketball Rebounding Robot — *Sep. 2025 – Present*

- Integrated ZED visual SLAM (VSLAM) and OpenCV on a Jetson Orin Nano for autonomous navigation
- Built a hierarchical ROS 2 state machine for autonomous target tracking, ball acquisition, and launching sequences
- Designed a custom flywheel system and x-drive chassis in Fusion 360 and handled all fabrication and assembly

Technical Skills

Languages & Tools: C++, Python, Java, OpenCV, SAM 3, Docker, Git, Linux (Ubuntu), Fusion 360 (CAD)
Computer Vision: 3D reconstruction, camera calibration, sensor fusion
Robotics & Simulation: ROS 2, RViz, Behavior Trees, Isaac Sim, Isaac Lab, MuJoCo, URDF, MJCF, USD, RL (PPO)
Hardware: LiDAR, ZED stereo camera, IMU, GPS, Jetson Orin Nano, Teensy 4.1, Boston Dynamics Spot, Jackal UGV